

Safety Data Sheet

1. Identification of the Substances/Preparation and of the Company/Undertaking

Product Name: General purpose polystyrene(GP525)

Other name: GPS;GPPS;Polystyrene resin,general purpose

Product property: High impact-resistant plastic

Formula: $[-CH(C_6H_5)CH_2-]_n$

End Uses: Used to make toys, household products, plastic parts, packaging containers, and disposable tableware, etc.

Manufacturer: NINGBO UNION KING NEW MATERIAL LIMITED

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2. Hazards Identification

GHS Hazard Category:

Classification according to Rules for classification and labeling of chemicals series standards GB 30000.2 ~ GB 30000.29-2013: Not classified as dangerous substance.

Labeling Elements

Pictogram(s):

Signal Word: Warning: None.

Hazard Statement(s): None.

Prevention Precautionary Statement(s): None.

Response Precautionary Statements: None.

Storage Precautionary Statements: None.

Disposal Precautionary Statements: None.

Physicochemical hazard: Combustible meet with open fire. May release hazardous gas or fumes.

Health hazard: None.

Environmental hazards: None.

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3. Composition information on Ingredients

Chemical properties:

Formula: $[-CH(C_6H_5)CH_2-]_n$

Substance	CAS #	Weight %
Polystyrene(PS)	9003-53-6	100

4. First-Aid Measures

4.1 First-aid measures

Eye Contact: Immediately rinse with flowing water or normal saline, occasionally lifting the upper and lower eyelids. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical help if you feel unwell.

Skin Contact: Remove contaminated cloth and wash thoroughly with running water. If skin irritation or a rash occurs: Get medical help if you feel unwell.

Inhalation: Move affected person to fresh air if irritation persists.

Ingestion: Give enough of water. Induce vomiting. Get medical advice/attention.

4.2 Acute or chronic effects:

The polymer is basically non-toxic. The toxicity is related to the non-polymerized monomer styrene of the polymer, which mainly has a strong irritating effect on the respiratory tract. Workers exposed to this product may cause symptoms such as pharyngitis and chronic tonsillitis pharyngitis. Some workers may cause dermatitis.

4.3 Indication of Any Immediate Medical Attention and Special Treatment needed

Advice to the rescuers: Transfer the patient to safe area. Consult a doctor. Show this chemical safety data sheet to the doctor.

Notes to Physician: Treat symptomatically.

5. Fire-Fighting Measures

Suitable Extinguishing Media: Extinguish with water spray, foam, dry chemical, carbon dioxide, sand.

Unsuitable Extinguishing Media: High volume water jet.

Special Hazards Arising from the Chemical:

Special Hazards: Combustion meets with open fire. May release hazardous gas or fumes.

Hazardous combustion products:

Carbon monoxide, carbon dioxide, nitrogen oxides, etc.

Special Protective Equipment for Fire-fighters: Fire-fighters should wear self-contained breathing apparatus with a positive pressure and full fire-fighting turn-out gear.

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Extinguishing the fire from upwind.

Transfer the containers from the fire to open place if possible.

Use water spray to cool the containers.

The fire fighting wastewater must have been harmless treatment before discharge into the sewage system.

6. Emergency Release Measures

Personal Precautions, Protective equipment and Emergency Procedures:

Provide enough ventilation. Immediately evacuate person to safe place, away from the leakage area and stay upwind. Emergency personnel shall use appropriate protective equipment when cleaning up leakage. See item 8 for details. Do not touch the leakage directly. Cut off the source of leaks if possible.

Precautions to Protect the Environment:

Avoid entry into the underground water system.

Methods and Materials for Containment and Cleaning Up:

Avoid creating dusty conditions. Remove away nearby sources of ignition. Vacuum or carefully scoop up spilled material and place in an appropriate container for recycle or transport to waste disposal sites.

7. Handling and Storage

Handling: *Use at well-ventilated area. Avoid contact with eyes and skin. Avoid formation of dust. Provide appropriate exhaust ventilation at places where dust is formed. Normal measures for preventive fire protection. Ensure the containers are closed if not used. Precautionary Statement(s) see item 2. Do not eat, drink or smoke at working place. Wash thoroughly after handling.*

Storage: *store in a cool, dry, well-ventilated place. Keep away from fire and heat. Keep container tightly closed. Store away from strong oxidizing agents. Store locked up.*

8. Exposure Controls/Personal Protection

Control parameters:

Exposure Limit(s): *GBZ 2.1-2019 Occupational exposure limits for hazardous agents in the workplace: Not listed.*

Biological Limit Values (BLVs): *Not established.*

Monitoring Method: *None.*

Appropriate Engineering Controls:

Not need special ventilation equipment. Provide enough natural ventilation. Facilities storing or utilizing this material should be equipped with an eyewash facility and a safety shower.

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Individual protective measure:

Respiration Protection: No special requirements with normal use. If dust exceeds recommended exposure limits, select proper respirator based on particle size of dust generated.

Hands Protection: Wear cotton gloves, or rubber coated gloves.

Skin Protection: Wear general working clothing.

Eye Protection: Wear safety goggles.

Hygiene Measures: Do not eat, drink or smoke at working place. Wash thoroughly after handling. Wash contaminated clothing before reuse.

9. Physical and Chemical Properties

Appearance:	White particles
Odor:	Odorless
Odor threshold value:	N/A
pH:	Neutral
Melting/ Freezing Point:	150°C~180°C
Boiling Point:	N/A
Flash Point:	N/A
Evaporation Rate:	N/A
Flammable:	Non-flammable solid
Explosion Limits in Air :	Not explosive
Vapor Pressure:	N/A
Vapor Density (Air=1):	N/A
Gravity (g/cm³):	1.04~1.09
Solubility :	Insoluble in water and polar solvents, soluble in organic solvents
Partition coefficient(n-octanol/ water):	N/A
Auto-combustion Temperature:	N/A
Decomposition Temperature:	>300°C
Viscosity:	N/A

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10. Stability and Reactivity

Stability: Stable when stored and used under normal ambient temperatures.

Hazardous Reactions: Violent reaction with strong oxidizing agents and increase vessel pressure.

Conditions to Avoided: High heat, open fire.

Incompatible Materials: Strong oxides, organic solvents.

Hazardous Decomposition Products: In case of high temperature, the following can be released: Carbon monoxide, carbon dioxide and small molecular hydrocarbons.

11. Toxicological Information

Acute Toxicity: Relatively Harmless.

Polystyrene (CAS#9003-53-6): $TD_{LO}=200\text{mg/kg}$ (Intravenously, Rat).

Skin Corrosion/Irritation: No skin irritation.

Serious Eye Damage/Eye Irritation: May cause mild mechanical irritation.

Respiratory or Skin Sensitization: No data available.

Germ Cell Mutagenicity: No data available.

Carcinogenicity: None of the ingredients is listed as carcinogen or suspected carcinogen by IARC, NTP, US OSHA or the European Directive (67/548/EEC).

Reproductive Toxicity: No data available.

Specific Target Organ Toxicity - Single Exposure: May cause respiratory irritation.

Specific Target Organ Toxicity - Repeated Exposure: No data available.

Aspiration Hazard: No aspiration hazard.

12. Ecological Information

Ecotoxicity: Harmless to aquatic life.

Polystyrene (CAS # 9003-53-6): $LC_{50}>500\text{ mg/L/48 h}$ (fish)

Persistence/Degradability: This product is not easily biodegradable and will exist in the environment for a long time.

BOD: 1%~3%

Bioaccumulation: The bioaccumulation of this product is not the most important environmental process.

BCF: 58~821

Mobility: insoluble in water, low migration, and not volatile.

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Other Adverse Effects: *Not known significant effects or critical hazards.*

13. Disposal Considerations

Waste Disposal Property: *Not dangerous waste.*

Waste Disposal Methods:

Recycle if possible. Products that cannot be recycled should be sent to a licensed professional waste disposal service to dispose of this material. Suggest incineration disposal

Contaminated containers may still present chemical hazard when empty. Return to supplier for recycling if possible.

Observe all federal, state, and local environmental regulations.

14. Transport Information

According to IATA, IMO/IMDG, ADR/RID

UN No: *N/A*

Proper Name: *Non-dangerous goods (non-restricted goods)*

Hazard Class: *N/A*

Packing Group: *N/A*

Environmental hazards: *Marine Pollutant: No*

Packing method: *Package according to the method recommended by the manufacturer.*

Such as plastic bags, corrugated cardboard boxes, corrugated cardboard boxes outside and film bags inside.

Attention in bulk transportation: *The package should be complete and sealed. Loading should be secure. Ensure the container does not leak, topple down, fall or damaged during the transportation. Should prevent sunshine, rain, heat. Prohibited transport with strong base, oxidants, food or food additives. Transport vehicles should be equipped with emergency treatment equipment. Road transport should be travel in according to the prescribed route. Do not stay in residential and congested areas.*

15. Regulatory Information

The following laws, regulations, rules and standards have made corresponding provisions about the safe use, storage, transportation, loading and unloading, classification and marking of the chemical:

Rules for classification and labeling of chemicals series standards GB 30000.2 ~ GB 30000.29-2013: *Not listed.*

Occupational Disease Prevention Law of the people's Republic of China:

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Classification catalogue of occupational hazard factors: Not listed.

Regulations on safety management of hazardous chemicals:

List of hazardous chemicals 2015 Edition (amended by 300th [2022] of Emergency Department in 2022): Not listed.

List of dangerous goods (GB 12268-2012): Not listed.

National list of hazardous wastes (2021 Edition): Not listed.

List of highly toxic substances (2003 Edition): Not listed.

Identification of major hazard installations for hazardous chemicals (GB18218-2018): Not listed.

List of Precursor Chemicals (The amended Edition of 2018) (amended by 58th [2021] of The State Council office in 2021): Not listed.

Environmental management measures for new chemical substances:

Inventory of Existing Chemical Substances in China (IECSC): Listed.

16. Other Information

Reference Documents:

- (1) Inventory of Existing Chemical Substances in China (IECSC)*
- (2) List of hazardous chemicals 2015 Edition (amended by 300th [2022] of Emergency Department in 2022)*
- (3) List of high toxic chemicals (2003 Edition)*
- (4) Identification of major hazard installations for hazardous chemicals (GB18218-2018)*
- (5) National list of hazardous wastes (2021 Edition)*
- (6) Regulations on the administration of precursor chemicals (The amended Edition of 2018) (amended by 58th [2021] of The State Council office in 2021)*
- (7) Classification catalogue of occupational hazard factors (2015 Edition)*
- (8) Occupational exposure limits for hazardous agents in the workplace --Part 1: Chemical hazardous agents (GBZ 2.1-2019)*
- (9) Safety data sheet for chemical products--Content and order of sections (GB16483-2008)*
- (10) Rules for classification and labelling of chemicals series standards (GB 30000.2 ~ GB 30000.29-2013)*
- (11) Guidance on the compilation of safety data sheet for chemical products (GB/T 17519-2013)*

Abbreviation:

CAS# - Chemical Abstract Service Number

LD50 - Lethal Dose 50

Further Information:

· This information is based on our present knowledge. However, this shall not constitute a guarantee for any specific product features and shall not establish a legally valid contractual relationship.

· This safety data sheet was prepared in accordance with GB16483-2008, GB/T 17519-2013 and GB

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30000.2~30000.29-2013.

Disclaimer:

The information in this SDS applies only to the product specified and, unless specified, not for the mixture of this product and other substances. This SDS provides only information on product use safety for those appropriately professionally trained users of the product. The user of this SDS must make an independent judgment on the applicability of the SDS under special use conditions. The writer of this SDS will not be responsible for any injuries caused from the use of this SDS.

Department Issuing SDS: NINGBO UNION KING NEW MATERIAL LIMITED

Issue Date: July 27, 2023

This SDS Service is provided by LANDER INSPECTION & CERTIFICATION CO., LTD.

Signed for and on behalf of

Lander Testing Lab

Xuanhong Yu

Chemistry Lab Engineer

Date: July 27, 2023

*****End*****

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